



# Test and Reliability

## Exercise Sheet 1

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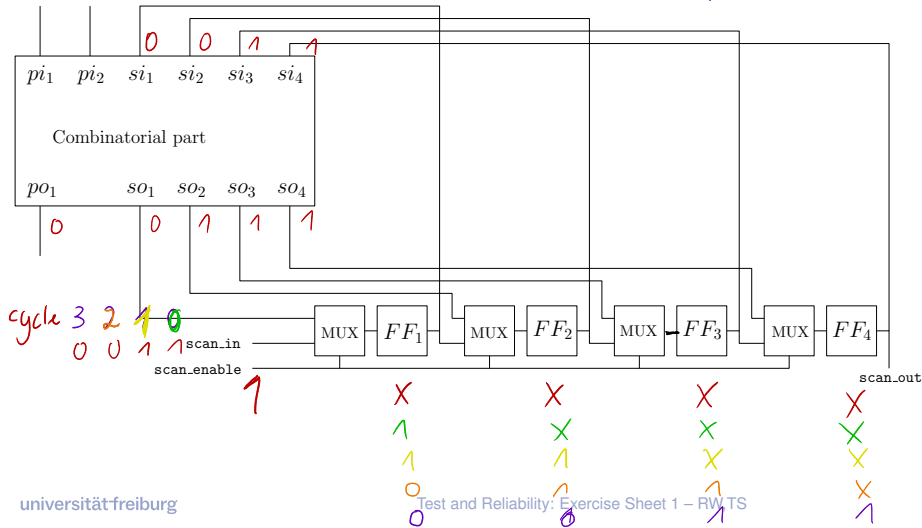


# Exercise 1

scan enable = 0

↳  $FF_1 = 0, FF_2 = 1, FF_3 = 1, FF_4 = 1$

↳ scan enable = 1 → "shift it out"

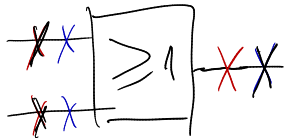
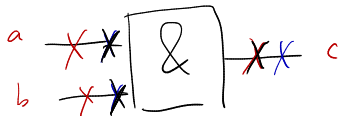


## Exercise 1

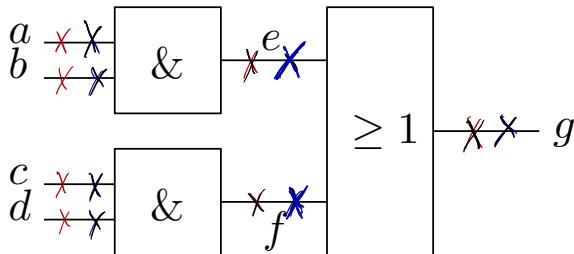
| Clock cycle | $pi_1$   | $pi_2$   | scan_enable | scan_in | $po_1$ | scan_out |
|-------------|----------|----------|-------------|---------|--------|----------|
| 1           | X        | X        | 1           | 1       | X      | 0        |
| 2           | X        | X        | 1           | 1       | X      | 0        |
| 3           | X        | X        | 1           | 0       | X      | 0        |
| 4           | X        | X        | 1           | 0       | X      | 1        |
| 5           | 1        | 1        | <u>0</u>    | X       | 0      | 1        |
| 6           | X        | X        | 1           | 1       | X      | 1        |
| 7           | X        | X        | 1           | 0       | X      | 1        |
| 8           | X        | X        | 1           | 1       | X      | 1        |
| 9           | X        | X        | 1           | 0       | X      | 0        |
| 10          | <u>0</u> | <u>1</u> | <u>0</u>    | X       | 1      | 0        |
| 11          | X        | X        | 1           | X       | X      | 0        |
| 12          | X        | X        | 1           | X       | X      | 0        |
| 13          | X        | X        | 1           | X       | X      | 1        |
| 14          | X        | X        | 1           | X       | X      | 1        |

~> 0111

## Exercise 2 a

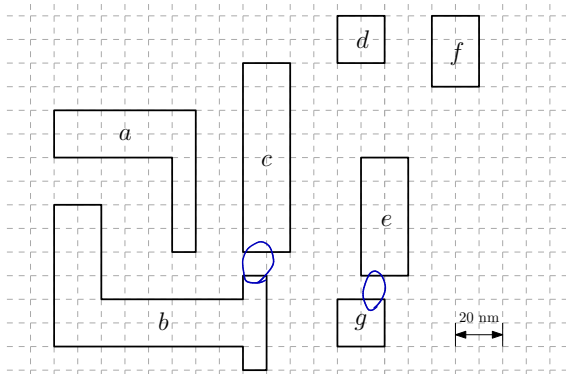
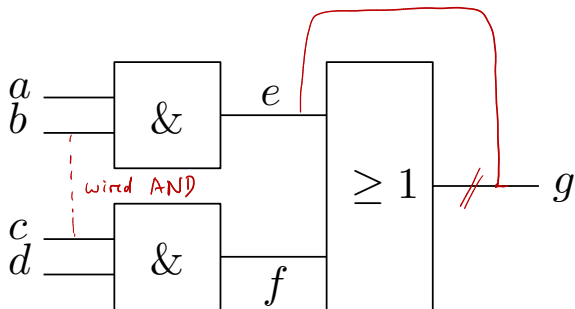


| a | b | good | a@0 | a@1 | b@0 | b@1 | c@0 | c@1 |
|---|---|------|-----|-----|-----|-----|-----|-----|
| 0 | 0 | 0    | 0   | 0   | 0   | 0   | 0   | 1   |
| 0 | 1 | 0    | 0   | 1   | 0   | 0   | 0   | 1   |
| 1 | 0 | 0    | 0   | 0   | 0   | 1   | 0   | 1   |
| 1 | 1 | 1    | 0   | 1   | 0   | 1   | 0   | 1   |



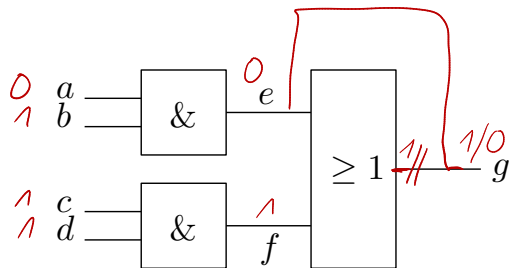
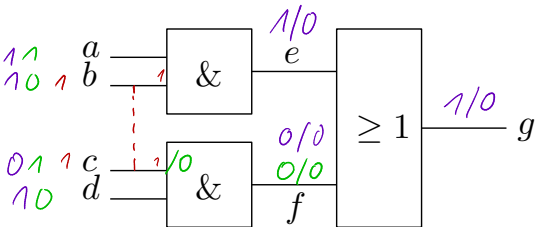
## Exercise 2 b)

$$\begin{array}{c} a \\ \text{---} \\ b \end{array} \Rightarrow \text{3D-E}$$

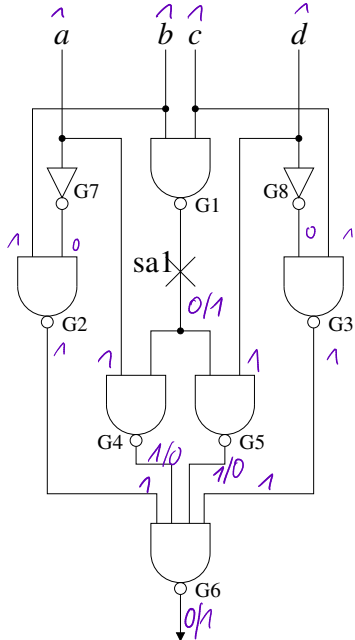


## Exercise 2 c)

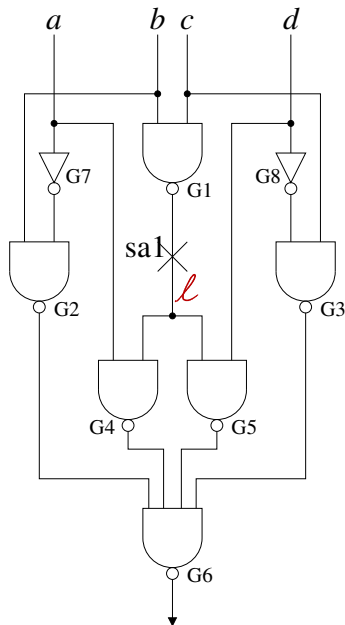
$\begin{matrix} a & b & c & d \\ 0 & 1 & 0 & 1 \end{matrix}$ 
 $\begin{matrix} 1 & 1 & 1 & 0 \end{matrix}$ 
 $\begin{matrix} 1 & 0 & 1 & 0 \end{matrix}$ 
 $\begin{matrix} 0 & 1 & 1 & 1 \end{matrix}$



## Exercise 3 a) b)



## Exercise 3 c)



slow-to-fall

↑  $t$     ↑  $t+\delta$

two tests:

1. set  $l$  to 1  
XXOX<sub>l</sub> or X<sub>l</sub>XX =
2. test for  $l @ 1$   
see b)